coat results in the rumination of endosperm in *Passiflora*. In Annonaceae and Aristolochiaceae definite ingrowth or infolding of the seed coat produces ruminate endosperm. The irregular surface of the seed coat makes endosperm ruminate in *Myristica*.

Functions of endosperm:

- It is the nutritive tissue for the developing embryo.
- In majority of angiosperms, the zygote divides only after the development of endosperm.
- Endosperm regulates the precise mode of embryo development.

Endosperm haustoria

Another interesting feature of the endosperm is the presence of haustoria. In the case of helobial endosperm the chalazal chamber itself acts as a haustorial structure. In cellular and nuclear endosperm special structures are produced towards the micropylar, chalazal, both micropylar and chalazal which may be in lateral direction depending on the species. These absorb nutrients from other outer tissue or from ovary tissue and supply them to the growing embryo.



Coconut milk is a basic nutrient medium which induces the differentiation of embryo (embryoids) and plantlets from various plant tissues. Coconut water from tender coconut is free-nuclear endosperm and white kernel part is cellular.

Embryogenesis

Development of Dicot embryo

The development of Dicot embryo (*Capsella bursa-pastoris)* is of Onagrad or crucifer type. The embryo develops at micropylar end of embryo sac.

The Zygote divides by a transverse division forming **upper or terminal cell** and **lower or basal cell**. The basal cell divides transversely and the terminal cell divides vertically to form a four celled proembryo. A second vertical division right angle to the first one takes place in terminal cell forming a four celled stage called **quadrant.** A transverse division in the quadrant results in eight cells arranged in two tiers of four each called **octant stage.**

The upper tier of four cells of the octant is called **epibasal** or anterior octant and the lower tier of four cells constitute **hypobasal** or posterior octants. A periclinal division in the octants results in the formation of 16 celled stage with eight cells in the outer and eight in the inner.

The outer eight cells represent the **dermatogen** and undergoes anticlinal division to produce epidermis. The inner eight cells divide by vertical and transverse division to form outer layer of **periblem** which give rise to cortex and a central region of **pleurome** which forms stele

During the development, the two cells of the basal cell undergoes several transverse division to form a six to ten celled **suspensor.** The embryo at this stage become **globular** and the suspensor helps to push the embryo deep into the endosperm. The uppermost cell of the suspensor enlarge to form a **haustorium**. The lowermost cell of the suspensor is called **hypophysis**. A transverse division and two vertical division right angle to each other of hypophysis results in the formation of eight cells. The eight cells are arranged in two tiers of four cells each The upper tier give rise to root cap and epidermis. At this stage embryo proper appears heart shaped, cell divisions in the **hypocotyl** and cotyledon regions of the embryo proper results in elongation. Further development results in curved horse shoe shaped embryo in the embryo sac. The mature embryo has a **radicle,** two **cotyledons** and a **plumule**(Figure 1.22).

Activity

Collect the fruits of *Tridax* (Cypsella). Using a needle dissect out the content, separate the embryo and observe different stages of dicot embryo – globular, torpedo, heart shaped under a dissection microscope.

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